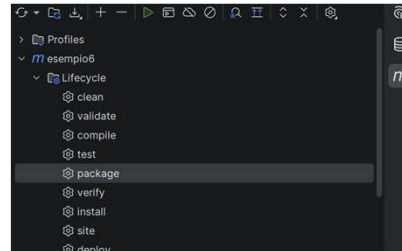


Web app deployment step-by-step

1. Generate the .jar:
 - Open the Maven tab in IJ:
 - Select LifeCycle
 - Click twice on package
2. Go in the project folder:
 - Go in the target folder
 - There is a – SNAPSHOT.jar



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Web app deployment step-by-step

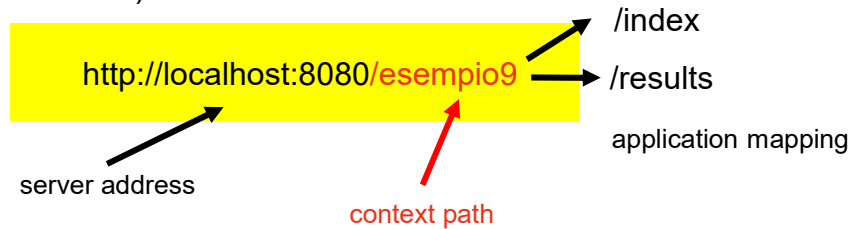
1. «Simulate» the cloud:
 - Copy the .jar in the folder you like (just to simulate the upload to the cloud)
 - Open the terminal of your OS
 - Go to the folder and run:


```
java -jar name_app_SNAPSHOT.jar
```
 - Wait Tomcat starts
 - Open the browser and type `http://localhost:8080/home`

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Deploying more complex web apps

- The context path of a web app is the base URL prefix under which the application's resources (e.g. controller, html pages and so on) are accessible



- To avoid issues during the deployment, we have to build URL in the proper way: do not hardcode them!

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The problem and the solution

- The problem is how the browser resolves relative URLs:

When a relative URL starts with a "/", the browser interprets it as being relative to the root of the domain (the part before the first / after the protocol and domain name)

`http://localhost:8080/compute`

- Thymeleaf has attributes designed to handle URL generation while being aware of the web application's context path :
`th:action`, `th:ref`, `th:src`, and so on

- `@{...}` syntax within the `th:`
 - "read" the context path
 - It automatically adds it at the beginning"

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<https://belief-driven-design.com/thymeleaf-part-1-basics-3a1d9/>

A more complex example...

Our first web Calculator

esempio9bis

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More than one Controller?

- We can use more than a Controller in our web app
- Exemplary cases (some of them):
 - Separation of functionalities
 - Different HTTP methods
 - User roles and permissions (admin vs. user; public vs. private)

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Le'ts come back to our calculator: We can do better!

- We can improve our calculator by applying what we studied about beans and dependency injection
- Try to separate the business logic (computation) from the code that manages HTTP requests and responses. Use the `@Autowired` annotation.
- For the sake of ease, you do not need anymore to notify the user about the operation that was performed

esempio10

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@Service annotations

- We know the `@Controller` annotation, a specialization of `@Component`
- To solve the exercise, we need to introduce `@Service`, another annotation that specializes `@Component`

	@Controller	@Service
Layer	Web layer	Business Logic
Purpose	Managing HTTP requests and responses (view or data)	Encapsulating business logic and service operations
Usage	Typically used in classes responsible for managing web requests, e.g. handling user input, and returning data or views in response to HTTP requests.	Typically used to represent business logic or any service-layer operations no necessarily related to HTTP requests e.g. processing data

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Let's do it all together!

- Sometimes, a HTTP response is prepared by more methods of the same Controller or (rarely) by more methods belonging to different Controllers.
- Each method is responsible to write a part of the response or to produce intermediate results to generate the final response



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The forward prefix

- Forward refers to the process of forwarding a request from one method (controller) to another one within the same web app
- It is an internal server-side routing mechanism performed by the Dispatcher Servlet
- It doesn't involve a new HTTP request from the client

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Summary

- Forward is a mechanism in which the request (and the Model) is passed from one method / Controller to another without the client being aware
- Consequences:
 - The URL in the browser does not change
 - The forwarding method and the forwarded method must be annotated both as `@GetMapping` or `@PostMapping` (the request is the same!)
- Avoid to use forward to a different Controller or Service

esempio11

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Working together to build the view

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Include

- Include is a mechanism to embed the content of another resource (e.g. a HTML fragment) inside the current resource.
- The included content is processed as part of the current response, but the original request isn't interrupted.
- Thymeleaf takes care to include content from another template or fragment using:

`th:include`

It inserts the fragment inside the tag where it's applied: the original element stays in the final output and contains the fragment content.

`th:replace`

It replaces the entire tag where it's applied with the fragment content, removing the tag and inserting only the fragment.

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Example

Index.html

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
  <title>Home Page</title>
</head>
<body>
  <div th:include="~{fragment.html}"></div>
  <div>
    <h2>Welcome to the home page!</h2>
  </div>
</body>
</html>
```

~{...} indicates that Thymeleaf has to look for the header in the templates folder

fragment.html

```
<header>
  <h1>Hello world!</h1>
</header>
```

What is included?

```
<div>
  <header>...</header>
</div>
```

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Example

Index.html

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
  <title>Home Page</title>
</head>
<body>
  <div th:replace="~{fragment.html}"></div>
  <div>
    <h2>Welcome to the home page!</h2>
  </div>
</body>
</html>
```

fragment.html

```
<header>
  <h1>Hello world!</h1>
</header>
```

What is included?

```
<header>...</header>
```

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esempio12

Ask somebody else: the redirect prefix

- Sometimes a Controller decides that it does not want to manage a request at all, so it forces the client to send a new request to somebody else
- Through redirect the server instructs the client's browser to make a new HTTP request to a different URL to the same/another web app in the same /another web server
- It is hidden to the client but:
 - There is a first response to the request with a status code 301 and a "location" header with a URL as the value
 - The browser sees 301 and looks for the "location" header
 - The browser makes a new request using the URL in the header in the previous response. The client might notice that the URL in the browser bar changed

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